

Toumi Adem

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SUMMARY

Engineering student specializing in **Artificial Intelligence and Data Science**, with a solid foundation in **Machine Learning, Deep Learning, NLP, Software Engineering, and Data Science**. Passionate about building intelligent solutions, developing mobile and web applications, and working with modern tools to solve real-world problems.

EDUCATION

École Supérieure en Informatique de Sidi Bel Abbès

Engineer's Degree in Artificial Intelligence and Data Science

Sidi Bel Abbès, Algeria

2021 – Expected 2026

TECHNICAL SKILLS

Programming Languages: Python, Dart, Java, TypeScript

Mobile Development: Flutter

Backend: Node.js (Express), FastAPI, GraphQL, SQL, MongoDB

Machine Learning: Scikit-learn, TensorFlow, PyTorch (Basics)

Tools: Git, Docker, Trello

EXPERIENCE

Intern – IT Department

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Sept 2024

Biskra, Algeria

- Explored applications of Artificial Intelligence for industrial monitoring and presented findings to engineers.
- Supported database administration tasks (SQL queries, performance tuning) and assisted in network administration.
- Contributed to automating data backup scripts and monitoring processes, improving reliability.
- Collaborated with engineers using Git for version control and Jira/Trello for task tracking.
- Observed CI/CD pipeline practices for deploying internal applications, gaining exposure to Docker-based workflows.

Backend Developer (Academic Project)

Hajj Management Platform

2023 – 2024

Sidi Bel Abbès, Algeria

- Developed a full-stack web platform using Node.js, TypeScript, React, and Next.js to manage pilgrim registration and logistics.
- Designed and optimized relational schemas, implemented authentication, and integrated REST APIs with FastAPI services.
- Collaborated in a team of students using Git and Trello to coordinate tasks and ensure timely delivery.

Research Project

Real-Time Monitoring and Anomaly Detection

2024

Sidi Bel Abbès, Algeria

- Designed a data pipeline for real-time monitoring of system metrics using Prometheus and Grafana.
- Built and trained a hybrid VAE-LSTM model in PyTorch for anomaly detection in time-series data.
- Integrated alerts into live dashboards to provide proactive notifications for abnormal system behavior.
- Optimized deployment workflows using Docker to ensure reproducibility and scalability.

PROJECTS

Hajj Management Platform

Academic Project

- Developed a web platform using Node.js, TypeScript, React, and Next.js to manage pilgrim registration and logistics.
- Designed database schemas and implemented user authentication system.

Object Detection and Tracking

Academic Project

- Explored object detection models including YOLOv8/YOLOv11, implemented dataset preparation and training pipeline.
- Performed comparative analysis of CSRT tracking and YOLO built-in tracking approaches.

Paraphrase Detection System

Mini Project - NLP

- Built a Siamese neural network with shared weights for semantic similarity detection.
- Implemented Bi-LSTM layers and multi-head attention to capture sequence context and dependencies.

Movie Recommendation System

Mini Project

- Developed a basic recommender system leveraging collaborative filtering techniques to suggest movies.

Real-Time Monitoring and Anomaly Detection

Deep Learning Project

- Designed and implemented a data pipeline for real-time monitoring of CPU/GPU metrics using Prometheus and Grafana.
- Built and trained a hybrid VAE-LSTM model in PyTorch for anomaly detection on time-series data.
- Integrated the anomaly detection model with live data streams to provide proactive alerts on abnormal system behavior.
- Optimized deployment and persistence of models, ensuring compatibility with production environments.

LANGUAGES

Arabic (Native)

French (Professional)

English (Professional)